SEAT No.________

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SARDAR PATEL UNIVERSITY

M. Sc. (Integrated) Biotechnology – Seventh Semester Examination

Thursday, 7th January, 2021

10:00 a.m. to 12:00 p.m.

PS07EIBM21: Plant Biotechnology

Note: 1) Figures to the right indicate marks

2) Draw diagram wherever necessary

Total marks: 70

Q – 1 (A) Choose the most appropriate alternative for the following: **(08)**

1. Who discovered cellular totipotency?
   - a) Haberlandt
   - b) Richard Roberts
   - c) Hamilton Smith
   - d) None of these
2. PTC is a redefined method of
   - a) Vegetative propagation
   - b) Asexual reproduction
   - c) Hybridization
   - d) Sexual reproduction
3. Chloroplast transformation requires
   - a) An efficient selection for transplastome
   - b) Suitable DNA delivery method
   - c) Chloroplast specific expression vector
   - d) All of these
4. _____ was the first higher plant to achieve chloroplast transformation.
   - a) Potato
   - b) Tobacco
   - c) Rice
   - d) None of these
5. _____ is required for the production of secondary metabolites.
   - a) Cell suspension
   - b) Callus
   - c) Protoplast
   - d) Meristem
6. _____ is not a class of secondary metabolite.
   - a) Amino acids
   - b) Terpenes
   - c) Phenolics
   - d) Alkaloids
7. A toxin produced by the Bt is coded by a gene named as
   - a) Cry protein
   - b) Bacillus
   - c) cry
   - d) Cyr
8. Which of the following gene is transferred in *Arabidopsis thaliana* for the production of polyhydroxybutyrate (biodegradable thermoplastic)?
   - a) Cyclodextrin glucosyl transferase
   - b) Mannitol dehydrogenase
   - c) Acetyl CoA reductase
   - d) RUBISCO

Q – 1 (B) Do as directed. **(16)**

1. Define callus.
2. Naturally grown plants shows somaclonal variations. (True/ False)
3. Any part of the plant/ tissue used for *In vitro* culturing is known as ______.
4. Micropropagation is carried out to produce somatic hybrids only. (True/ False)
5. Write any two limitations of chloroplast transformation.

6. High level of gene expression is obtained through chloroplast transformation. (True/ False)
7. Which transformation methods are widely used for chloroplast transformation?
8. What are transplastomic plants?
9. Secondary metabolites are not directly needed by the plant. (True/ False)
10. Mention the methods used for selection of cell lines producing higher amount of metabolites.
11. _____ gave the method of cell aggregate cloning.
12. Give two applications of secondary metabolites.
13. Cry proteins have 5 functional domains. (True/ false)
14. Glyphosate inhibits _____ enzyme.
15. What is subunit vaccine?
16. Dextran enhances vir gene expression. (True/ False)

Q – 2 Answer any seven in short. **(14)**

1. Mention the stages of micropropagation.
2. List the factors influencing somaclonal variation.
3. Mention the advantages of chloroplast transformation.
4. Draw the neat and labelled diagram of chloroplast.
5. Which are the main problems associated with secondary metabolite production?
6. Enlist main approaches used for secondary metabolite production.
7. List advantages of cell culture system over the conventional cultivation of whole plant for the production of secondary metabolites.
8. Give schematic representation of production of toxin fragment from Cry proteins.
9. Why genetic engineering approach is better than conventional breeding for development of disease resistant plants?

Q – 3 Discuss advantages and disadvantages of micropropagation. **(08)**

OR

Q – 3 Write a note on somaclonal variation. **(08)**

Q – 4 Describe the biolistic method of gene transfer in detail. **(08)**

OR

Q – 4 Explain the process of chloroplast transformation through *Agrobacterium.* **(08)**

Q – 5 What are secondary metabolites? Describe various types of secondary metabolites in detail. **(08)**

OR

Q – 5 Discuss the factors affecting secondary metabolite production. **(08)**

Q – 6 Give an account on edible vaccines. **(08)**

OR

Q – 6 Write a note on development of drought resistant plants. **(08)**

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